

CO2 ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

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① Responsive Web Design (RWD) and its importance:

Responsive Web Design (RWD) is a web development approach that allows a website to automatically adjust its layout, images, and content according to the screen size and device being used (mobile, tablet or desktop).

Importance:

- * Provides a consistent user experience across all devices.
- * Improves website accessibility and usability
- * Increases customer satisfaction and engagement.
- * Enhances SEO since search engines favour mobiles.

1. Fluid Layouts:

- * Use relative units like %, vw, vh, em and rem instead fixed pixels.

2. Flexible images: CSS

```
img {
```

```
max-width: 100%;
```

```
height: auto;
```

```
}
```

3. Media Queries: CSS

```
@media (max-width: 786px) {
```

```
body {
```

font size: 16px;

}

}

Using CSS Flexbox for Adaptive Layout: HTML

Flexbox automatically arranges items according to available space.

```
<div class="container">
```

```
<div class="box">Menu</div>
```

```
<div class="box">content</div>
```

```
<div class="box">sidebar</div>
```

CSS:-

container

display: flex;

gap: 10px;

}

box

flex: 1;

padding: 20px;

background: light blue;

}

@media (max-width: 768px) {

container {

flex-direction: column;

}

}

Improvements for Better user Experience:

* use responsive navigation menus.

* optimize image for faster loading.

* Make buttons larger font sizes.

Q2) Client-side validation and its purpose:

Client-side validation checks user input in the browser before sending data to the server.

Purpose:-

- * Prevents invalid data submission.
- * Provides immediate feedback.
- * Reduces unnecessary server requests.
- * Improves user experience.

JavaScript Event Handling in Validation:

JavaScript validation input using events such as:

- * submit
- * input
- * change
- * blur

Role of Regular Expressions:

Regular expressions (Regex) check whether input follows a required pattern.

Email Validation:

```
let Pattern = /^[^s@]+@^[^s@]+.[^s@]+$/;
```

```
if (Pattern.test(email)) {
```

```
    alert("Valid Email");
```

```
}
```

```
else {
```

```
    alert("Invalid Email");
```

```
}
```


Phone Number Validation: Java script:-

let phonePattern = /^[0-9]{10}\$/;

Example Form:- HTML

```
<form id="myForm">  
<input type="email" id="email">  
<button>submit</button>  
</form>
```

Improvements:-

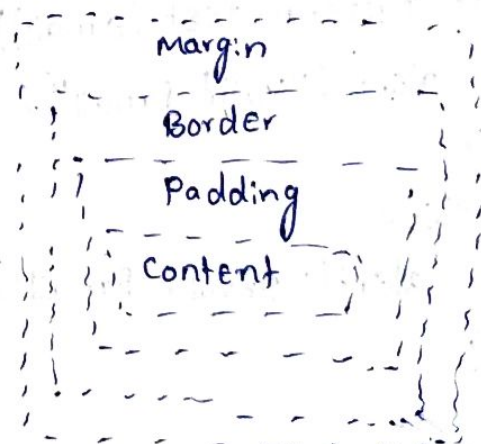
- * Display error messages beside input fields.
- * Use HTML5 validation (required, pattern, type="email").
- * Always perform server-side validation for security.
- * Sanitize user input to prevent attacks.

Q3 CSS Box Model and Layout issues:-

The CSS Box Model describes how elements are displayed on a webpage.

Components of model BOX:-

1. content - Actual text or image
2. padding - space around the content.
3. Border - surrounds the padding.
4. Margin - space outside the border.



CSS Positioning Techniques:

Static - Default position.

Relative - moves relative to its normal position.

Absolute - positioned relative to the nearest positioned ancestor.

Fixed - Fixed to the browser window.

Sticky - Acts like relative until scrolling.

causes of Layout Breaking:

* Fixed widths using pixels.

* Large image without scaling.

* incorrect positioning.

* overflowing content.

* Missing responsive design.

using Media Queries: CSS:

```
@media (max-width: 768px) {
```

```
  .container {
```

```
    flex-direction: column;
```

```
  }
```

```
}
```

Best Practices:

* use Flexbox or Grid * Apply box-sizing: border-box.

* use Percentage widths instead of fixed pixels.

CSS:-

```
* {
```

```
  box-sizing: border-box;
```

```
}
```

* optimize images

* Test on different devices.

24) Document object Model (DOM) and Dynamic content:

The Document object Model (DOM) is a Programming interface that represents an HTML document as a tree of objects.

JavaScript uses the DOM to access and modify webpage elements dynamically without reloading the page.

Role of DOM:-

- * Access HTML elements.
- * Modify text and styles.
- * Add or remove elements.
- * Respond to user actions.
- * update content dynamically.

DOM Manipulation Example: HTML

```
<p id="message">Welcome</p>
```

```
<button onclick="changeText()">click Here</button>
```

JavaScript:-

```
function changeText(){
```

```
    document.getElementById("message").innerHTML="Welcome  
    to JavaScript!";
```

```
}
```

Another Example (Adding a new item) HTML

```
<ul id="list">
```

```
<li>Apple</li>
```

```
</ul>
```

```
<button onclick="addItem()">Add</button>
```


Javascript:

```
function addItem(){
```

```
let li = document.createElement("li");
```

```
li.innerHTML = "orange";
```

```
document.getElementById("list").appendChild(li);
```

```
}
```

Improvements for performance and user Experience:

- * update only required DOM elements instead of reloading the page.

- * Minimize unnecessary DOM manipulations.

- * event delegation for better performance.

- * Display loading indicators during data fetching.

- * validate user input before updating the DOM.

- * write clean and modular Javascript code.